

Parth Bibekar

Curriculum Vitae

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Education

- Sept 2024–
Ongoing **Institute of Bioengineering, SV, EPFL, Switzerland,**
PhD candidate in Computational and Quantitative Biology.
Advisor: Prof. Matteo Dal Peraro
- 2019–2024 **Indian Institute of Science Education and Research, Kolkata, India,**
BSMS Dual Degree, Major in Biology & minor in Computational and Data Sciences.
CGPA – 8.21/10

Research Interests

Machine Learning for RNA design, RNA therapeutics, and Protein design

Current Position

- Sept 2024–
Ongoing I am a PhD student at the [Laboratory for Biomolecular modeling](#) at EPFL, Switzerland, under the supervision of Prof. Matteo Dal Peraro. Developing geometric deep learning and reinforcement learning techniques for various applications in biomolecular modeling, with a focus on RNA design, mRNA therapeutics, and modeling protein-nucleotide complexes.

Publications

- Jun 2025 Bibekar et al., Context-aware geometric deep learning for RNA sequence design. NeurIPS 2025 AI4D3 Workshop (2025), [10.1101/2025.06.21.660801](https://arxiv.org/abs/10.1101/2025.06.21.660801)
- Apr 2024 Bibekar et al., PeSTo-Carbs: Geometric Deep Learning for Prediction of Protein-Carbohydrate Binding Interfaces. Journal of Chemical Theory and Computation (2024), [10.1021/acs.jctc.3c01145](https://doi.org/10.1021/acs.jctc.3c01145)
- Feb 2023 Dutta et al., Conformational ensemble of the NSP1 CTD in SARS-CoV-2: Perspectives from the free energy landscape, Biophysical Journal (2023), [10.1016/j.bpj.2023.02.010](https://doi.org/10.1016/j.bpj.2023.02.010)

Work Experience

- May 2022 – **Research Intern, EPFL, Switzerland,**
Aug 2022 *with Prof. Matteo Dal Peraro at the Laboratory for Biomolecular Modeling.*
 - Developed geometric deep learning models for studying protein-carbohydrate interactions.
- Jun 2021 – **Research Intern, IISER Kolkata, India,**
Jan 2023 *with Prof. Neelanjana Sengupta at (bio)molecular Computations: Energetics and Dynamics group.*
 - Developed unsupervised ML methods for estimating biomolecular transition barriers from MD simulations.

Achievements

Aug 2023 **E3 Program Fellow.**

- I was awarded the EPFL Excellence in Engineering Program fellowship, funding a project in the Laboratory for Biomolecular Modeling at EPFL, Switzerland.

Jun 2022 **ThinkSwiss Research Scholarship.**

- I was awarded the ThinkSwiss Research Scholarship of for a period of three months for a research project at EPFL, Switzerland.

Dec 2021 **National HPC Hackathon.**

- Part of the winning team in the National HPC Hackathon organised by Amazon Web Services and Intel in association with the Office of Principal Scientific Advisor to the Government of India.

Teaching Experience

2026 **TA EPFL Team iGEM.**

- I am a supervisor for EPFL's [iGEM](#) team 2026. Supervising computational and mathematical modeling aspect for a synthetic biology project based on developing biosensors.

2025 **TA BIOENG-455: Computational Cell Biology.**

- Supervised students for Dissipative particle dynamics simulations.

2024, 2026 **TA BIO-315: Structural biology.**

- Supervised students for modules: Molecular Dynamics, Molecular Modeling and CryoEM analysis.

Conferences & Invited Talks

Mar 2026 Physics of Living Systems Seminar (**Talk**), EPFL, Switzerland

Jan 2026 EPFL AIxBIO seminar (**Talk**), EPFL, Switzerland

Jan 2026 Swiss RNA Workshop (**Poster**), Bern, Switzerland

Jan 2026 frontiers2026 symposium on NanoBioEngineering (**Talk**), EPFL, Switzerland

Jan 2026 SIB AI4Biomolecules SIG seminar (**Talk**), Bern, Switzerland

Dec 2025 NeurIPS AI4D3 Workshop (**Poster**), San Diego, USA

Oct 2025 European RosettaCon (**Poster**), Ljubljana, Slovenia

Oct 2025 Structure Based Drug Discovery (**Poster**), Sestri Levante, Italy

Nov 2024 European RosettaCon (**Poster**), Copenhagen, Denmark

Reviewing

2026 **CASP17 Assessor.**

- Independent assessor for the Ensembles category of the 17th Critical Assessment of Structure Prediction ([CASP17](#)).